

Project 3: SQL Data Analysis

DecodeLabs Industrial Training Kit | Data Analytics | Batch 2026

Orders	Customers	Gross revenue	Avg order value	Period
1,200	1,189	1,264,761.96	1,053.97	2023-01-01 to 2025-06-30

1. Objective

Use SQL to extract business insights from an e-commerce orders dataset. The project demonstrates writing **SELECT** queries, filtering with **WHERE**, sorting with **ORDER BY**, grouping with **GROUP BY**, and aggregating with **COUNT**, **SUM** and **AVG**, as required by the project brief. It also goes beyond the minimum with **HAVING** and percentage-contribution analysis, both suggested in the brief's conclusion.

Requirement (from brief)	Where it is covered
Write SELECT queries	Section 1 (queries 1.1 - 1.5)
WHERE (equality, comparison, AND/OR, IN, BETWEEN, LIKE)	Section 2 (queries 2.1 - 2.6)
ORDER BY	Section 3 (3.1 - 3.3), and used throughout
COUNT, SUM, AVG (plus MIN, MAX)	Sections 4 and 5
GROUP BY	Section 5 (5.1 - 5.8)
HAVING (brief's conclusion tip)	Section 6 (6.1 - 6.3)
Percentage contribution (brief's conclusion tip)	Query 7.1
Execution order and the 'Alias Trap'	Query 7.5

2. Dataset

Source: **cleaned_dataset.xlsx**, loaded into a SQLite table named **orders** (1,200 rows, 14 columns). The Date column is stored as text in YYYY-MM-DD format.

Column	Type	Description
OrderID	TEXT (PK)	Unique order identifier
Date	TEXT	Order date, YYYY-MM-DD
CustomerID	TEXT	Customer identifier
Product	TEXT	Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet
Quantity / UnitPrice	INTEGER / REAL	Units ordered and price per unit
ShippingAddress	TEXT	Delivery address
PaymentMethod	TEXT	Online, Cash, Credit Card, Debit Card, Gift Card
OrderStatus	TEXT	Cancelled, Returned, Pending, Shipped, Delivered
TrackingNumber	TEXT	Shipment tracking ID
ItemsInCart	INTEGER	Items in the cart at checkout
CouponCode	TEXT	FREESHIP, SAVE10, WINTER15 or No Coupon
ReferralSource	TEXT	Instagram, Email, Google, Facebook, Referral
TotalPrice	REAL	Order total (= Quantity x UnitPrice)

Data quality checks (all passed): no duplicate OrderIDs (query 1.5), no NULLs in the columns checked (query 4.2), and TotalPrice equals Quantity x UnitPrice on every row (query 1.4).

3. How to reproduce

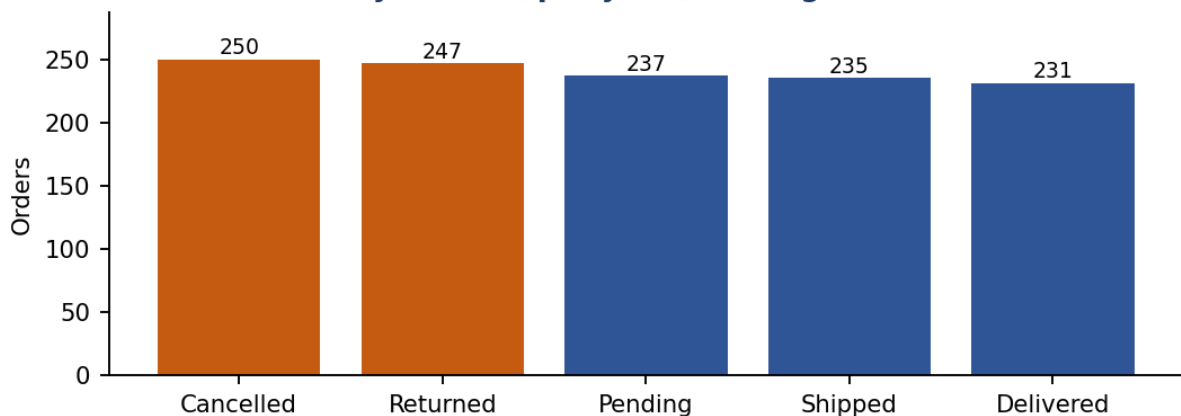
1. Install DB Browser for SQLite and open **database/orders.db**.
2. Go to the **Execute SQL** tab and open **sql/Project3_SQL_Queries.sql**.
3. Highlight one query and press Ctrl+Enter to run only that query.

For MySQL or PostgreSQL, import data/cleaned_dataset.xlsx (saved as CSV) into a table named orders and replace strftime() as noted at the top of the .sql file.

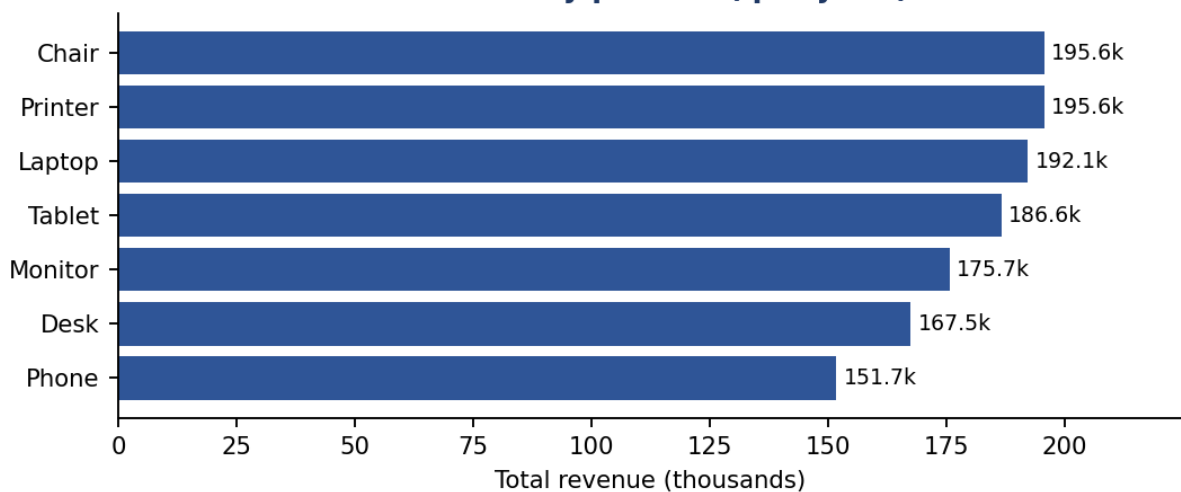
4. Key insights

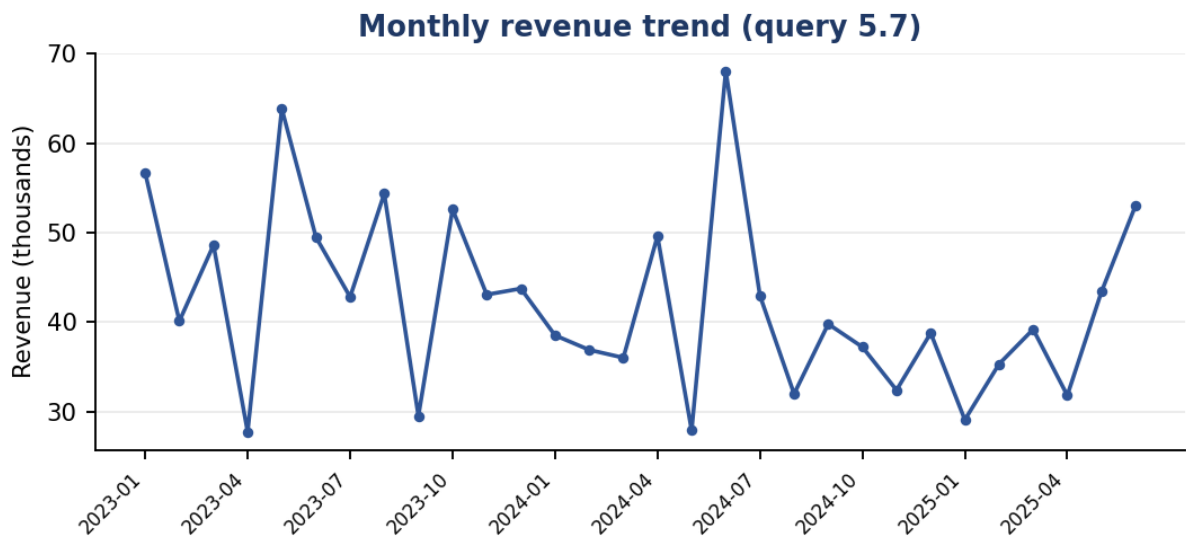
Theme	Finding
Scale	1,200 orders from 1,189 customers between 2023-01-01 and 2025-06-30; gross revenue 1,264,761.96, average order value 1,053.97. (4.1)
Order outcomes	Only 231 of 1,200 orders (19.2%) are Delivered. Cancelled (250) + Returned (247) = 497 orders, or 41.4% of all orders - status is spread almost evenly across all five values. (5.1)
Revenue at risk	Excluding Cancelled and Returned orders, net revenue is 745,088.05 versus gross 1,264,761.96 (58.9% retained). Note this still includes Pending and Shipped orders. (7.3)
Products	Revenue is spread fairly evenly: Chair leads with 15.47% of total and Phone is lowest with 12.00%. Monitor has the highest cancel/return rate (43.56%). (5.2, 7.1, 7.2)
Payment & channel	Credit Card has the highest average order value (1,127.55). Instagram is the top referral source by total revenue, while Email leads on Delivered revenue only. (5.3, 5.4, 7.4)
Coupons	Average order value differs only between 1,035.90 and 1,070.41 across coupon groups, so coupons show no clear order-size uplift in this data. (5.5)
Trend	Yearly revenue: 2023 = 552,643.24, 2024 = 480,235.87, 2025 = 231,882.85. 2025 only covers Jan-Jun, so compare like-for-like: H1 2023 = 286,501.52, H1 2024 = 257,059.34, H1 2025 = 231,882.85. (5.6, 5.7)
Customers	Only 11 customers appear more than once (all with exactly 2 orders), so repeat purchasing is very low. (6.1)
Data quality	No duplicate OrderIDs, no NULLs in any checked column, and TotalPrice equals Quantity x UnitPrice on every row. (1.4, 1.5, 4.2)

Orders by status (query 5.1) - orange = lost orders



Revenue by product (query 5.2)





5. Queries and results

Each query shows the business question, the SQL, and the result (first 10 rows; the full output is in results/Project3_SQL_Analysis.xlsx).

Section 1: SELECT basics

1.1 Preview the table

Question: What does the raw data look like?

Skills: SELECT, LIMIT

```
SELECT *
FROM orders
LIMIT 10;
```

Rows returned: 10 (showing first 5)

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2,853.10
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.70
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2,753.40
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2,504.04

1.2 Choose specific columns + alias

Question: Show only the columns needed for a sales report.

Skills: SELECT, AS (alias)

```
SELECT OrderID AS order_id,
       Date AS order_date,
       Product AS product,
       Quantity AS qty,
       TotalPrice AS order_value
FROM orders
LIMIT 10;
```

Rows returned: 10

order_id	order_date	product	qty	order_value
ORD200000	2023-01-04	Monitor	5	2,853.10
ORD200001	2024-08-23	Phone	2	302.70
ORD200002	2024-02-27	Tablet	5	2,753.40
ORD200003	2023-10-15	Chair	1	273.19
ORD200004	2025-05-08	Printer	4	2,504.04
ORD200005	2023-10-23	Phone	2	491.72
ORD200006	2025-06-17	Laptop	1	664.42
ORD200007	2023-05-12	Monitor	5	747.75
ORD200008	2025-04-02	Phone	2	268.56
ORD200009	2023-11-21	Desk	4	2,037.52

1.3 Distinct values

Question: Which products do we sell?

Skills: SELECT DISTINCT, ORDER BY

```
SELECT DISTINCT Product
FROM orders
ORDER BY Product;
```

Rows returned: 7

Product
Chair
Desk
Laptop
Monitor
Phone
Printer
Tablet

1.4 Data-quality check: TotalPrice = Quantity x UnitPrice

Question: Is the TotalPrice column trustworthy? (0 rows = every order reconciles)

Skills: SELECT, WHERE, calculated column

```
SELECT OrderID,
       Quantity,
       UnitPrice,
       ROUND(Quantity * UnitPrice, 2) AS calculated_total,
       TotalPrice
FROM orders
WHERE ROUND(Quantity * UnitPrice, 2) <> ROUND(TotalPrice, 2);
```

Rows returned: 0 - an empty result is the expected outcome (no problems found).

1.5 Data-quality check: duplicate OrderIDs

Question: Does any OrderID appear more than once? (0 rows = all unique)

Skills: GROUP BY, HAVING, COUNT

```
SELECT OrderID, COUNT(*) AS times_seen
FROM orders
GROUP BY OrderID
HAVING COUNT(*) > 1;
```

Rows returned: 0 - an empty result is the expected outcome (no problems found).

Section 2: WHERE filtering

2.1 Equality filter

Question: List all delivered orders.

Skills: WHERE, =

```
SELECT OrderID, Date, Product, Quantity, TotalPrice
FROM orders
WHERE OrderStatus = 'Delivered'
ORDER BY Date;
```

Rows returned: 231 (showing first 10)

OrderID	Date	Product	Quantity	TotalPrice
ORD200698	2023-01-04	Laptop	2	776.44
ORD201008	2023-01-13	Monitor	3	1,232.64
ORD201080	2023-01-20	Laptop	5	2,323.70
ORD200692	2023-01-23	Desk	5	984.20
ORD200829	2023-02-05	Monitor	1	76.93
ORD200100	2023-02-17	Phone	3	1,994.37
ORD200128	2023-02-17	Tablet	4	84.28
ORD200673	2023-02-21	Tablet	2	745.84
ORD200404	2023-03-09	Monitor	2	198.20
ORD200794	2023-03-09	Printer	4	1,539.40

2.2 Numeric comparison

Question: Which orders are worth 3,000 or more?

Skills: WHERE, >=

```
SELECT OrderID, Date, Product, Quantity, UnitPrice, TotalPrice
FROM orders
WHERE TotalPrice >= 3000
ORDER BY TotalPrice DESC;
```

Rows returned: 34 (showing first 10)

OrderID	Date	Product	Quantity	UnitPrice	TotalPrice
ORD200789	2023-08-17	Tablet	5	691.28	3,456.40
ORD201122	2023-06-07	Monitor	5	678.19	3,390.95
ORD200632	2023-05-02	Laptop	5	678.16	3,390.80
ORD200469	2023-11-26	Chair	5	676.98	3,384.90
ORD200328	2023-02-28	Tablet	5	674.04	3,370.20
ORD200107	2023-03-27	Printer	5	670.75	3,353.75
ORD200326	2024-07-01	Laptop	5	670.48	3,352.40
ORD201065	2023-10-30	Printer	5	666.80	3,334.00
ORD201031	2023-02-28	Phone	5	664.51	3,322.55
ORD200463	2023-05-26	Laptop	5	662.78	3,313.90

2.3 Combining conditions (AND)

Question: Which delivered orders are high-value (2,000+)?

Skills: WHERE, AND

```
SELECT OrderID, Date, Product, PaymentMethod, TotalPrice
FROM orders
WHERE OrderStatus = 'Delivered'
      AND TotalPrice >= 2000
ORDER BY TotalPrice DESC;
```

Rows returned: 38 (showing first 10)

OrderID	Date	Product	PaymentMethod	TotalPrice
ORD200789	2023-08-17	Tablet	Online	3,456.40
ORD200632	2023-05-02	Laptop	Gift Card	3,390.80
ORD201065	2023-10-30	Printer	Debit Card	3,334.00
ORD200361	2024-06-29	Printer	Credit Card	3,299.25
ORD200837	2025-06-14	Chair	Online	3,277.75
ORD201079	2024-06-20	Phone	Gift Card	3,170.00
ORD200511	2024-03-15	Monitor	Cash	2,876.20
ORD200578	2024-10-10	Monitor	Credit Card	2,830.35
ORD200883	2025-05-13	Printer	Online	2,807.40
ORD200769	2023-07-06	Printer	Cash	2,714.20

2.4 IN + BETWEEN

Question: Which 2024 orders were for Laptops, Phones or Tablets?

Skills: WHERE, IN, BETWEEN, AND

```
SELECT OrderID, Date, Product, TotalPrice
FROM orders
WHERE Product IN ('Laptop', 'Phone', 'Tablet')
      AND Date BETWEEN '2024-01-01' AND '2024-12-31'
ORDER BY Date;
```

Rows returned: 195 (showing first 10)

OrderID	Date	Product	TotalPrice
ORD200737	2024-01-03	Phone	2,131.48
ORD200082	2024-01-05	Tablet	2,260.80
ORD200399	2024-01-07	Tablet	2,119.65
ORD201105	2024-01-08	Tablet	1,427.07
ORD201164	2024-01-08	Tablet	1,577.85
ORD200674	2024-01-09	Laptop	1,435.00
ORD200712	2024-01-09	Tablet	136.80
ORD200781	2024-01-10	Phone	2,621.30
ORD200476	2024-01-15	Phone	1,027.18
ORD200624	2024-01-16	Tablet	1,709.80

2.5 Pattern matching (LIKE)

Question: Which orders used a 'SAVE' coupon (SAVE10)?

Skills: WHERE, LIKE, wildcard %

```
SELECT OrderID, Date, Product, CouponCode, TotalPrice
FROM orders
WHERE CouponCode LIKE 'SAVE%'
ORDER BY Date;
```

Rows returned: 286 (showing first 10)

OrderID	Date	Product	CouponCode	TotalPrice
ORD200645	2023-01-02	Laptop	SAVE10	300.10
ORD200000	2023-01-04	Monitor	SAVE10	2,853.10
ORD201003	2023-01-11	Printer	SAVE10	1,599.85
ORD200058	2023-01-15	Phone	SAVE10	1,898.76
ORD200383	2023-01-20	Chair	SAVE10	1,575.20
ORD200057	2023-01-22	Printer	SAVE10	1,571.35
ORD200019	2023-01-23	Monitor	SAVE10	1,120.00
ORD200556	2023-01-28	Laptop	SAVE10	654.27
ORD200977	2023-01-28	Chair	SAVE10	2,597.88
ORD200022	2023-01-29	Monitor	SAVE10	99.24

2.6 Negation (<>) and OR

Question: Which orders were lost (Cancelled OR Returned)?

Skills: WHERE, OR, IN

```
SELECT OrderID, Date, Product, OrderStatus, TotalPrice
FROM orders
WHERE OrderStatus = 'Cancelled'
   OR OrderStatus = 'Returned'
ORDER BY TotalPrice DESC;
```

Rows returned: 497 (showing first 10)

OrderID	Date	Product	OrderStatus	TotalPrice
ORD201122	2023-06-07	Monitor	Returned	3,390.95
ORD200469	2023-11-26	Chair	Cancelled	3,384.90
ORD200328	2023-02-28	Tablet	Cancelled	3,370.20
ORD200326	2024-07-01	Laptop	Returned	3,352.40
ORD200527	2024-08-19	Chair	Cancelled	3,267.35
ORD200768	2023-09-04	Tablet	Cancelled	3,267.30
ORD200889	2024-10-27	Monitor	Cancelled	3,253.60
ORD200802	2024-02-17	Chair	Cancelled	3,223.20
ORD200957	2023-01-12	Monitor	Returned	3,219.45
ORD200086	2024-06-19	Printer	Cancelled	3,215.15

Section 3: ORDER BY sorting

3.1 Top 10 orders by value

Question: What are our 10 biggest orders?

Skills: ORDER BY DESC, LIMIT

```
SELECT OrderID, Date, Product, Quantity, TotalPrice
FROM orders
ORDER BY TotalPrice DESC
LIMIT 10;
```

Rows returned: 10

OrderID	Date	Product	Quantity	TotalPrice
ORD200789	2023-08-17	Tablet	5	3,456.40
ORD201122	2023-06-07	Monitor	5	3,390.95
ORD200632	2023-05-02	Laptop	5	3,390.80
ORD200469	2023-11-26	Chair	5	3,384.90
ORD200328	2023-02-28	Tablet	5	3,370.20
ORD200107	2023-03-27	Printer	5	3,353.75
ORD200326	2024-07-01	Laptop	5	3,352.40
ORD201065	2023-10-30	Printer	5	3,334.00
ORD201031	2023-02-28	Phone	5	3,322.55
ORD200463	2023-05-26	Laptop	5	3,313.90

3.2 Multi-column sort

Question: Sort by product A-Z, and within each product by value (highest first).

Skills: ORDER BY with two columns

```
SELECT Product, OrderID, TotalPrice
FROM orders
ORDER BY Product ASC, TotalPrice DESC
LIMIT 20;
```

Rows returned: 20 (showing first 10)

Product	OrderID	TotalPrice
Chair	ORD200469	3,384.90
Chair	ORD200837	3,277.75
Chair	ORD200527	3,267.35
Chair	ORD200802	3,223.20
Chair	ORD200241	3,078.35
Chair	ORD200876	3,044.55
Chair	ORD200222	2,773.44
Chair	ORD200963	2,651.70
Chair	ORD200287	2,620.72
Chair	ORD200707	2,600.16

3.3 Most recent orders

Question: What are the 10 latest orders?

Skills: ORDER BY DESC on a date

```
SELECT OrderID, Date, Product, OrderStatus, TotalPrice
FROM orders
ORDER BY Date DESC
LIMIT 10;
```

Rows returned: 10

OrderID	Date	Product	OrderStatus	TotalPrice
ORD200256	2025-06-30	Chair	Returned	379.05
ORD201107	2025-06-30	Tablet	Cancelled	126.42
ORD200176	2025-06-28	Tablet	Pending	1,723.32
ORD200773	2025-06-28	Phone	Cancelled	1,024.66
ORD200882	2025-06-28	Phone	Delivered	548.24
ORD201039	2025-06-28	Desk	Cancelled	598.46
ORD200226	2025-06-27	Desk	Delivered	1,764.15
ORD200064	2025-06-25	Printer	Cancelled	391.83
ORD200055	2025-06-24	Printer	Cancelled	1,033.20
ORD200163	2025-06-24	Tablet	Cancelled	641.39

Section 4: Aggregations

4.1 Overall KPIs

Question: How big is the business overall?

Skills: COUNT, SUM, AVG, MIN, MAX

```
SELECT COUNT(*)           AS total_orders,
       COUNT(DISTINCT CustomerID) AS unique_customers,
       SUM(Quantity)        AS units_sold,
       ROUND(SUM(TotalPrice), 2) AS gross_revenue,
       ROUND(AVG(TotalPrice), 2) AS avg_order_value,
       MIN(TotalPrice)        AS smallest_order,
       MAX(TotalPrice)        AS largest_order,
       MIN(Date)              AS first_order_date,
       MAX(Date)              AS last_order_date
FROM orders;
```

Rows returned: 1

total_orders	unique_customers	units_sold	gross_revenue	avg_order_value	smallest_order	largest_order	first_order_date	last_order_date
1,200	1,189	3,535	1,264,761.96	1,053.97	11.39	3,456.40	2023-01-01	2025-06-30

4.2 COUNT(*) vs COUNT(column)

Question: Does any column contain NULLs? (PDF 'Crucial Rule': COUNT(*) counts every row, COUNT(col) skips NULLs)

Skills: COUNT(*), COUNT(col), NULL handling

```
SELECT COUNT(*)           AS all_rows,
       COUNT(CouponCode) AS non_null_coupon,
       COUNT(TotalPrice)  AS non_null_total,
       COUNT(CustomerID)  AS non_null_customer
FROM orders;
```

Rows returned: 1

all_rows	non_null_coupon	non_null_total	non_null_customer
1,200	1,200	1,200	1,200

Section 5: GROUP BY

5.1 Orders by status

Question: How many orders are in each status?

Skills: GROUP BY, COUNT, ORDER BY alias

```
SELECT OrderStatus,  
       COUNT(*) AS order_count  
FROM orders  
GROUP BY OrderStatus  
ORDER BY order_count DESC;
```

Rows returned: 5

OrderStatus	order_count
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

5.2 Product performance

Question: Which product brings in the most revenue?

Skills: GROUP BY, COUNT, SUM, AVG

```
SELECT Product,  
       COUNT(*) AS orders,  
       SUM(Quantity) AS units_sold,  
       ROUND(SUM(TotalPrice), 2) AS total_revenue,  
       ROUND(AVG(TotalPrice), 2) AS avg_order_value  
FROM orders  
GROUP BY Product  
ORDER BY total_revenue DESC;
```

Rows returned: 7

Product	orders	units_sold	total_revenue	avg_order_value
Chair	178	562	195,620.11	1,098.99
Printer	181	542	195,612.61	1,080.73
Laptop	173	535	192,126.56	1,110.56
Tablet	179	497	186,568.95	1,042.28
Monitor	163	480	175,651.41	1,077.62
Desk	170	508	167,459.93	985.06
Phone	156	411	151,722.39	972.58

5.3 Payment method analysis

Question: Which payment method is most used, and which has the highest average order?

Skills: GROUP BY, COUNT, AVG

```
SELECT PaymentMethod,  
       COUNT(*) AS orders,  
       ROUND(SUM(TotalPrice), 2) AS total_revenue,  
       ROUND(AVG(TotalPrice), 2) AS avg_order_value  
FROM orders  
GROUP BY PaymentMethod  
ORDER BY orders DESC;
```

Rows returned: 5

PaymentMethod	orders	total_revenue	avg_order_value
Online	258	262,442.94	1,017.22
Cash	246	259,786.29	1,056.04
Credit Card	234	263,847.63	1,127.55
Debit Card	232	232,361.18	1,001.56
Gift Card	230	246,323.92	1,070.97

5.4 Marketing channel analysis

Question: Which referral source drives the most revenue?

Skills: GROUP BY, COUNT, SUM, AVG

```
SELECT ReferralSource,
       COUNT(*)           AS orders,
       ROUND(SUM(TotalPrice), 2) AS total_revenue,
       ROUND(AVG(TotalPrice), 2) AS avg_order_value
FROM orders
GROUP BY ReferralSource
ORDER BY total_revenue DESC;
```

Rows returned: 5

ReferralSource	orders	total_revenue	avg_order_value
Instagram	259	275,285.45	1,062.88
Email	250	261,808.55	1,047.23
Google	241	250,441.48	1,039.18
Facebook	228	250,410.90	1,098.29
Referral	222	226,815.58	1,021.69

5.5 Coupon effectiveness

Question: Do coupons change order size?

Skills: GROUP BY, COUNT, SUM, AVG

```
SELECT CouponCode,
       COUNT(*)           AS orders,
       ROUND(SUM(TotalPrice), 2) AS total_revenue,
       ROUND(AVG(TotalPrice), 2) AS avg_order_value,
       ROUND(AVG(ItemsInCart), 2) AS avg_items_in_cart
FROM orders
GROUP BY CouponCode
ORDER BY avg_order_value DESC;
```

Rows returned: 4

CouponCode	orders	total_revenue	avg_order_value	avg_items_in_cart
FREESHIP	313	335,036.99	1,070.41	5.53
SAVE10	286	304,840.02	1,065.87	5.46
No Coupon	309	322,401.41	1,043.37	5.56
WINTER15	292	302,483.54	1,035.90	5.38

5.6 Yearly revenue

Question: How does revenue compare year to year?

Skills: GROUP BY on a date expression (strftime)

```
SELECT strftime('%Y', Date) AS year,
       COUNT(*)           AS orders,
       ROUND(SUM(TotalPrice), 2) AS total_revenue
FROM orders
GROUP BY strftime('%Y', Date)
ORDER BY year;
```

Rows returned: 3

year	orders	total_revenue
2023	510	552,643.24
2024	459	480,235.87
2025	231	231,882.85

5.7 Monthly revenue trend

Question: What does the month-by-month revenue trend look like?

Skills: GROUP BY on a date expression, ORDER BY

```
SELECT strftime('%Y-%m', Date) AS year_month,  
       COUNT(*) AS orders,  
       ROUND(SUM(TotalPrice), 2) AS total_revenue  
FROM orders  
GROUP BY strftime('%Y-%m', Date)  
ORDER BY year_month;
```

Rows returned: 30 (showing first 10)

year_month	orders	total_revenue
2023-01	47	56,685.75
2023-02	37	40,117.66
2023-03	43	48,609.37
2023-04	31	27,751.71
2023-05	49	63,836.84
2023-06	45	49,500.19
2023-07	44	42,820.66
2023-08	51	54,352.14
2023-09	29	29,526.67
2023-10	47	52,607.85

5.8 Two-column GROUP BY

Question: How many orders does each product have in each status?

Skills: GROUP BY two columns

```
SELECT Product,  
       OrderStatus,  
       COUNT(*) AS orders  
FROM orders  
GROUP BY Product, OrderStatus  
ORDER BY Product, orders DESC;
```

Rows returned: 35 (showing first 10)

Product	OrderStatus	orders
Chair	Cancelled	45
Chair	Pending	41
Chair	Delivered	33
Chair	Shipped	31
Chair	Returned	28
Desk	Pending	38
Desk	Cancelled	35
Desk	Shipped	33
Desk	Returned	32
Desk	Delivered	32

Section 6: HAVING

6.1 Repeat customers

Question: Which customers placed more than one order?

Skills: GROUP BY, HAVING, COUNT

```
SELECT CustomerID,  
       COUNT(*)           AS orders,  
       ROUND(SUM(TotalPrice), 2) AS total_spent  
FROM orders  
GROUP BY CustomerID  
HAVING COUNT(*) > 1  
ORDER BY orders DESC, total_spent DESC;
```

Rows returned: 11 (showing first 10)

CustomerID	orders	total_spent
C38840	2	5,723.23
C97593	2	2,855.22
C98474	2	2,175.34
C70659	2	1,853.96
C94569	2	1,675.35
C46651	2	1,360.89
C35852	2	1,248.39
C14847	2	1,097.81
C91155	2	704.82
C21191	2	647.77

6.2 Filter groups by an aggregate

Question: Which months earned more than 50,000 in revenue?

Skills: GROUP BY, HAVING SUM

```
SELECT strftime('%Y-%m', Date) AS year_month,  
       ROUND(SUM(TotalPrice), 2) AS total_revenue  
FROM orders  
GROUP BY strftime('%Y-%m', Date)  
HAVING SUM(TotalPrice) > 50000  
ORDER BY total_revenue DESC;
```

Rows returned: 6

year_month	total_revenue
2024-06	68,068.54
2023-05	63,836.84
2023-01	56,685.75
2023-08	54,352.14
2025-06	53,047.40
2023-10	52,607.85

6.3 WHERE + GROUP BY + HAVING together

Question: Among DELIVERED orders only, which products average more than 1,000 per order?

Skills: WHERE (row filter) then HAVING (group filter)

```
SELECT Product,
       COUNT(*)           AS delivered_orders,
       ROUND(AVG(TotalPrice), 2) AS avg_order_value
FROM orders
WHERE OrderStatus = 'Delivered'
GROUP BY Product
HAVING AVG(TotalPrice) > 1000
ORDER BY avg_order_value DESC;
```

Rows returned: 5

Product	delivered_orders	avg_order_value
Printer	29	1,312.23
Monitor	31	1,161.28
Tablet	28	1,135.52
Phone	38	1,061.72
Laptop	40	1,017.86

Section 7: Business insights

7.1 Percentage contribution by product

Question: What share of total revenue does each product contribute? (PDF conclusion suggests this)

Skills: GROUP BY, SUM, scalar subquery, percentage

```
SELECT Product,
       ROUND(SUM(TotalPrice), 2) AS product_revenue,
       ROUND(100.0 * SUM(TotalPrice) / (SELECT SUM(TotalPrice) FROM orders), 2) AS pct_of_total
FROM orders
GROUP BY Product
ORDER BY product_revenue DESC;
```

Rows returned: 7

Product	product_revenue	pct_of_total
Chair	195,620.11	15.47
Printer	195,612.61	15.47
Laptop	192,126.56	15.19
Tablet	186,568.95	14.75
Monitor	175,651.41	13.89
Desk	167,459.93	13.24
Phone	151,722.39	12.00

7.2 Cancellation & return rate by product

Question: Which product loses the most orders to cancellations or returns?

Skills: GROUP BY, SUM(CASE WHEN ...), rate calculation

```
SELECT Product,
       COUNT(*) AS total_orders,
       SUM(CASE WHEN OrderStatus = 'Cancelled' THEN 1 ELSE 0 END) AS cancelled,
       SUM(CASE WHEN OrderStatus = 'Returned' THEN 1 ELSE 0 END) AS returned,
       ROUND(100.0 * SUM(CASE WHEN OrderStatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END)
            / COUNT(*), 2) AS lost_order_pct
FROM orders
GROUP BY Product
ORDER BY lost_order_pct DESC;
```

Rows returned: 7

Product	total_orders	cancelled	returned	lost_order_pct
Monitor	163	35	36	43.56
Tablet	179	34	43	43.02
Laptop	173	35	39	42.77
Chair	178	45	28	41.01
Printer	181	35	38	40.33
Phone	156	31	31	39.74
Desk	170	35	32	39.41

7.3 Gross vs net revenue by product

Question: How much revenue is at risk? Net = orders that are NOT Cancelled or Returned.

Skills: GROUP BY, conditional SUM, CASE WHEN

```
SELECT Product,
       ROUND(SUM(TotalPrice), 2) AS gross_revenue,
       ROUND(SUM(CASE WHEN OrderStatus NOT IN ('Cancelled','Returned')
            THEN TotalPrice ELSE 0 END), 2) AS net_revenue,
       ROUND(SUM(CASE WHEN OrderStatus IN ('Cancelled','Returned')
            THEN TotalPrice ELSE 0 END), 2) AS revenue_lost
FROM orders
GROUP BY Product
ORDER BY net_revenue DESC;
```

Rows returned: 7

Product	gross_revenue	net_revenue	revenue_lost
Printer	195,612.61	120,998.36	74,614.25
Chair	195,620.11	120,320.97	75,299.14
Laptop	192,126.56	108,710.54	83,416.02
Tablet	186,568.95	103,413.13	83,155.82
Desk	167,459.93	99,040.75	68,419.18
Monitor	175,651.41	98,069.37	77,582.04
Phone	151,722.39	94,534.93	57,187.46

7.4 Top 3 referral sources by delivered revenue

Question: Which 3 marketing channels deliver the most completed (Delivered) revenue?

Skills: WHERE, GROUP BY, SUM, ORDER BY, LIMIT

```
SELECT ReferralSource,
       COUNT(*) AS delivered_orders,
       ROUND(SUM(TotalPrice), 2) AS delivered_revenue
FROM orders
WHERE OrderStatus = 'Delivered'
GROUP BY ReferralSource
ORDER BY delivered_revenue DESC
LIMIT 3;
```

Rows returned: 3

ReferralSource	delivered_orders	delivered_revenue
Email	60	64,553.52
Facebook	42	52,149.38
Instagram	52	51,299.56

7.5 The 'Alias Trap' done correctly (PDF slide 12)

Question: Filter on a computed value. Using the alias in WHERE fails because WHERE runs before SELECT; wrap it in a subquery instead.

Skills: Execution order, alias, subquery

```
-- WRONG in MySQL / PostgreSQL / SQL Server (error: column "rev" does not exist),
-- because WHERE runs before SELECT. (SQLite is lenient and allows it, so don't rely on that.)
--   SELECT Quantity * UnitPrice AS rev FROM orders WHERE rev > 3000;

-- RIGHT option A: repeat the expression inside WHERE
--   SELECT Quantity * UnitPrice AS rev FROM orders WHERE Quantity * UnitPrice > 3000;

-- RIGHT option B (used below): compute the alias in a subquery, filter it in the outer query
SELECT *
FROM (
    SELECT OrderID, Product, Quantity * UnitPrice AS rev
    FROM orders
) AS t
WHERE rev > 3000
ORDER BY rev DESC;
```

Rows returned: 34 (showing first 10)

OrderID	Product	rev
ORD200789	Tablet	3,456.40
ORD201122	Monitor	3,390.95
ORD200632	Laptop	3,390.80
ORD200469	Chair	3,384.90
ORD200328	Tablet	3,370.20
ORD200107	Printer	3,353.75
ORD200326	Laptop	3,352.40
ORD201065	Printer	3,334.00
ORD201031	Phone	3,322.55
ORD200463	Laptop	3,313.90

6. Conclusion

All requirements of the brief were met: 32 queries cover SELECT, WHERE, ORDER BY, GROUP BY and the COUNT, SUM and AVG aggregations, plus HAVING, CASE WHEN, subqueries and percentage contribution. The main business takeaway is that roughly four in ten orders are cancelled or returned, so net revenue is far lower than gross revenue; reducing cancellations and returns is a larger lever than promoting any single product, since revenue is spread evenly across all seven products and coupons show no clear effect on order size. Selected SQL results were independently re-computed with Excel formulas and matched on every check (see results/Project3_SQL_Analysis.xlsx, sheet 'Excel Cross-Check').